Area and Volumes of Revolution

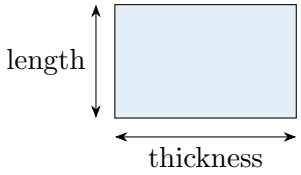
Visual Discussion Notes and Worked Solutions

MATH 182 | October 16, 2026 | Revised edition

Class format: 30 minutes. Use the central examples during class; the expanded derivations, extensions, and practice solutions support preparation and review. The two practice problems are unchanged.

1. One slice explains the formulas

Start with a bounded region. Draw one thin rectangle and identify its length and thickness. For area, add the rectangles. For volume, rotate them and add the resulting pieces.



Small area = length × thickness
Thin rectangles become exact in the limit.

The rectangle above is schematic: an actual integration slice is very thin.

$$A = \int_a^b [\text{top} - \text{bottom}]\, dx, \qquad A = \int_c^d [\text{right} - \text{left}]\, dy.$$

Choose the slice, then choose the formula

Method	Slice relative to axis	Small volume
Disk	Perpendicular; reaches axis	$\pi R^2\, dt$
Washer	Perpendicular; has a hole	$\pi(R^2 - r^2)\, dt$
Shell	Parallel	$2\pi r h\, dt$

Here dt is dx or dy. Radii are distances to the rotation axis. The shell height h is the length of the slice, even when that length is horizontal. **Setup routine:** sketch and shade → find bounds → draw a slice → label distances → write the integral → evaluate and check.

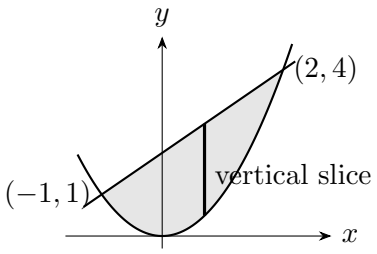
Teaching sequence

0–6 minutes	Area example: why top minus bottom?
6–15 minutes	Disk and washer examples: where is the hole?
15–24 minutes	Shell example: why circumference times height?
24–30 minutes	Practice setups and exit check.

2. Example 1. Area between a line and a parabola

Find the area enclosed by $y = x + 2$ and $y = x^2$.

Intersections: $x^2 = x + 2$ gives $(x - 2)(x + 1) = 0$, so $x = -1, 2$. The line lies above the parabola on this interval.



Step 1 — Find the intersections

The equation $x^2 = x + 2$ identifies the endpoints of the enclosed region. Its solutions are -1 and 2, so these become the x-limits.

Step 2 — Identify top and bottom

At $x = 0$, the line has height 2 and the parabola has height 0. Furthermore, $(x+2) - x^2 = (2-x)(x+1) \geq 0$ throughout $[-1, 2]$.

Step 3 — Measure the slice

The slice has height $(x + 2) - x^2$ and thickness dx. Therefore

$$A = \int_{-1}^{2} (x + 2 - x^2)\, dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2} = \frac{10}{3} - \left(-\frac{7}{6}\right) = \boxed{\frac{9}{2} \text{ square units}}.$$

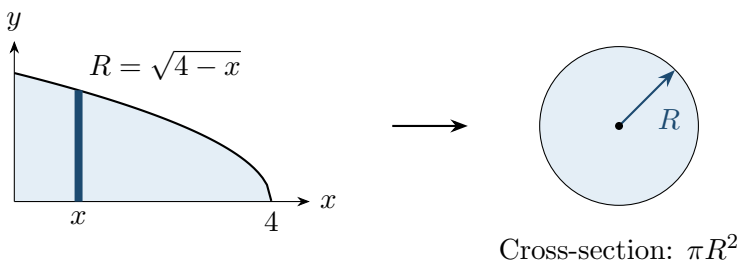
Check your understanding. Why must we subtract the parabola from the line? Because a slice length is top minus bottom and must be nonnegative.

Geometric intuition. The integral adds the vertical gaps between the curves. The bottom curve need not be the x-axis. Reversing the subtraction gives a negative signed integral, not the geometric area.

Prompt: "Point to the top and bottom of one slice before writing any formula."

3. Example 2: a slice becomes a disk

Problem. Rotate the region under $y = \sqrt{4 - x}$, above the x-axis, for $0 \le x \le 4$, about the x-axis.



Cross-section: πR^2

Step 1 — Understand the region

The curve passes through $(0, 2)$ and $(4, 0)$. The region lies between the curve and $y = 0$.

Step 2 — Choose the slice

The rotation axis is horizontal. A vertical slice is perpendicular to it, has thickness dx, and extends from the axis to the curve.

Step 3 — Measure the radius

The distance from $y = 0$ to $y = \sqrt{4 - x}$ is $R(x) = \sqrt{4 - x}$. There is no hole: $r = 0$.

Step 4 — Write the small volume and integral

$$dV = \pi \left(\sqrt{4 - x}\right)^2 dx = \pi(4 - x)\, dx, \qquad V = \pi \int_0^4 (4 - x)\, dx.$$

Step 5 — Evaluate

$$V = \pi \left[4x - \frac{x^2}{2}\right]_0^4 = \pi[(16 - 8) - 0] = \boxed{8\pi \text{ cubic units}}.$$

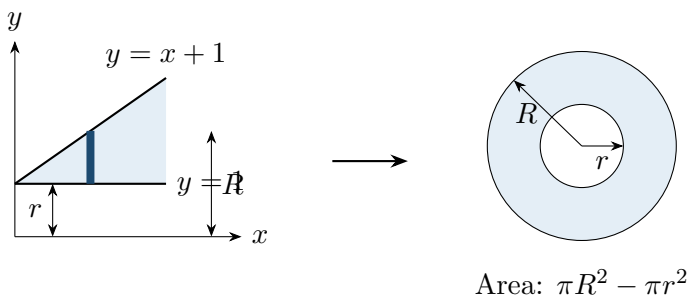
Geometric intuition. Imagine stacking thin coins with changing radii. The circular area is πR^2; multiplying by the small thickness produces volume. The square root disappears because the radius is squared.

Pause and check: Why does this problem produce a disk rather than a washer?

Answer: the representative slice reaches the axis; there is no central gap.

4. Example 3: a slice becomes a washer

Problem. Rotate the region between $y = x + 1$ and $y = 1$, for $0 \le x \le 2$, about the x-axis.



Area: $\pi R^2 - \pi r^2$

Step 1 — Choose a vertical slice

It is perpendicular to the horizontal rotation axis. It starts at $y = 1$ and ends at $y = x + 1$, so rotating it leaves a hole.

Step 2 — Measure both radii from the axis

$$R = (x + 1) - 0 = x + 1, \qquad r = 1 - 0 = 1.$$

The slice length is x, but the outer radius is $x + 1$. A radius starts at the axis, not at the bottom curve.

Step 3 — Subtract disk areas and integrate

$$\begin{aligned} V &= \pi \int_0^2 [(x+1)^2 - 1^2]\, dx \\ &= \pi \int_0^2 (x^2 + 2x + 1 - 1)\, dx = \pi \int_0^2 (x^2 + 2x)\, dx \\ &= \pi \left[\frac{x^3}{3} + x^2 \right]_0^2 = \pi \left(\frac{8}{3} + 4 \right) = \boxed{\frac{20\pi}{3}}. \end{aligned}$$

Geometric intuition. A washer is an outer disk with an inner disk removed. Subtract areas, not radii: $R^2 - r^2 \neq (R - r)^2$.

Shifted-axis extension (review). About $y = -2$, the same region has

$$R = (x + 1) - (-2) = x + 3, \qquad r = 1 - (-2) = 3.$$

$$V = \pi \int_0^2 [(x+3)^2 - 9]\, dx = \pi \left[\frac{x^3}{3} + 3x^2 \right]_0^2 = \boxed{\frac{44\pi}{3}}.$$

5. Cylindrical shells: parallel slices

A slice parallel to the rotation axis sweeps out a thin cylindrical shell. Unrolling it gives a rectangle with length equal to its circumference.

$$\boxed{dV = 2\pi(\text{radius})(\text{height})(\text{thickness})}.$$

The shell height is the length of the original slice. For a horizontal slice, this is a horizontal length.

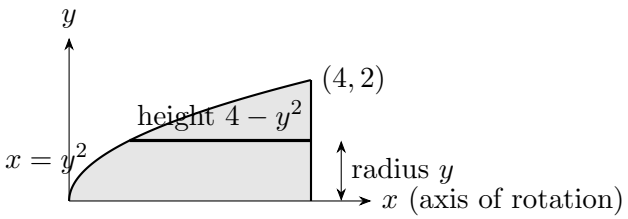
Example 4. Horizontal shells about the x-axis

Rotate the first-quadrant region bounded by $x = y^2$, $x = 4$, and $y = 0$ about the x-axis. Use shells.

Bounds: $y^2 = 4$ gives the upper height $y = 2$. Thus $0 \le y \le 2$.

One horizontal slice:

$$\text{radius} = y, \qquad \text{height} = \text{right} - \text{left} = 4 - y^2, \qquad \text{thickness} = dy.$$



Why the bounds are y-values. The slice moves upward from $y = 0$ to $y = 2$. For a horizontal slice, the thickness is dy; therefore its bounds must be heights.

Evaluate step by step:

$$V = 2\pi \int_0^2 y(4 - y^2)\, dy$$
$$= 2\pi \left[2y^2 - \frac{y^4}{4}\right]_0^2 = 2\pi\,[(8 - 4) - (0 - 0)] = \boxed{8\pi \text{ cubic units}}.$$

Independent check: vertical disks have radius $\sqrt{x}$ for $0 \le x \le 4$, so $V = \pi \int_0^4 x\, dx = 8\pi$.

Choose the direction before writing a formula

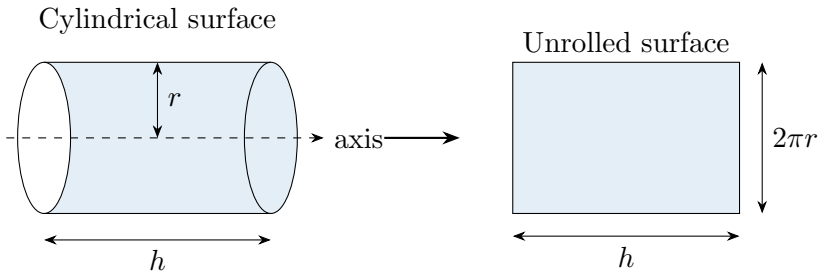
Axis	Vertical slices (dx)	Horizontal slices (dy)
Horizontal	Disks/washers	Shells
Vertical	Shells	Disks/washers

Vertical-shell model: about the y-axis, a region with top $f(x)$ and bottom $g(x)$, contained in $x \ge 0$, has

$$V = 2\pi \int_a^b x[f(x) - g(x)]\, dx.$$

Geometric intuition. Moving upward makes the shell radius larger but its horizontal length shorter. The integrand accounts for both changes. Do not square the shell height.

6. Why the shell formula works



The curved surface has area $(2\pi r)h$. Give it a small radial thickness Δr to approximate a shell volume:

$$\Delta V \approx (2\pi r)h\,\Delta r.$$

For a cylindrical shell with constant height h, the exact finite-thickness volume is

$$\begin{aligned}\Delta V &= \pi[(r+\Delta r)^2 - r^2]h \\ &= 2\pi r h\,\Delta r + \pi h(\Delta r)^2.\end{aligned}$$

The second term vanishes in the integral limit. When the slice length varies, its variation is also handled by the limit. Thus $dV = 2\pi r h\,dr$.

A comparison that prevents memorization

	Washer	Shell
What is added?	Circular cross-sections	Thin cylindrical layers
Slice direction	Perpendicular to axis	Parallel to axis
Distances	Outer and inner radius	Radius and slice length
Volume element	$\pi(R^2 - r^2)\,dt$	$2\pi r h\,dt$

Pause and check: Does rotation about the x-axis always require dx?

No. Washers use dx here, but horizontal shells use dy. The slice direction determines the differential. **Boundary check.** If the top, bottom, left, or right boundary changes formula within the interval, split the integral at that transition or choose another slice direction. A single formula must describe every slice where it is used.

7. Guided practice — unchanged problems

Practice A — washer setup. Rotate the region between $y = 3$ and $y = x$, for $0 \le x \le 3$, about the x-axis. Label R and r, then set up the volume integral.

Sketch, label distances, and write the integral:

Practice B — shell setup. Rotate the region under $y = 4 - x$ and above $y = 0$, for $0 \le x \le 4$, about the y-axis. Label the shell radius and height, then set up the volume integral.

Sketch, label distances, and write the integral:

Exit check

For rotation about the y-axis, explain why a vertical slice produces a shell and a horizontal slice produces a disk or washer. **Self-check:** axis marked; slice drawn; nonnegative distances; limits matched to dx or dy;

exact result; square units for area and cubic units for volume.

8. Practice solutions and geometric checks

Practice A: washers about the x-axis.

1. A vertical slice extends from $y = x$ to $y = 3$, for $0 \le x \le 3$.
2. Distances from $y = 0$ give $R = 3$ and $r = x$.
3. The washer area is $\pi(9 - x^2)$, so

$$V = \pi \int_0^3 (9 - x^2)\,dx = \pi \left[9x - \frac{x^3}{3}\right]_0^3 = \pi(27 - 9) = \boxed{18\pi}.$$

Geometric intuition. The solid is a cylinder of radius 3 and length 3 with a cone removed. The cylinder volume is 27π and the cone volume is $\frac{1}{3}\pi(3^2)(3) = 9\pi$. Their difference is 18π.

Practice B: shells about the y-axis.

1. Use vertical slices, parallel to the vertical rotation axis.
2. The radius is the distance from $x = 0$ to the slice: $r = x$.
3. The height is top minus bottom: $h = (4 - x) - 0 = 4 - x$.
4. Integrate over $0 \le x \le 4$:

$$\begin{aligned} V &= 2\pi \int_0^4 x(4 - x)\,dx = 2\pi \int_0^4 (4x - x^2)\,dx \\ &= 2\pi \left[2x^2 - \frac{x^3}{3}\right]_0^4 = 2\pi \left(32 - \frac{64}{3}\right) = \boxed{\frac{64\pi}{3}}. \end{aligned}$$

Geometric intuition. The solid is a cone with radius 4 and height 4. Its volume $\frac{1}{3}\pi(4^2)(4) = 64\pi/3$ independently confirms the shell calculation.

If you get stuck

Ask: Where is the axis? Which direction is the slice? What shape does it sweep out? Where do the radius and slice length begin and end? **Exit-check answer:** a vertical slice is parallel to the y-axis and sweeps out a cylindrical layer. A horizontal slice is perpendicular and sweeps out a circular cross-section, with or without a hole.